

Job Loss and Occupational Job Security: Evidence from Post-Displacement Sorting

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Abstract

Existing research on the cost of job loss has often overlooked the role of job sorting following displacement in contributing to recursive unemployment. This study estimates the effect of displacement on the job security associated with the occupations matched by displaced workers following dismissal. Using a dataset containing the employment histories of approximately four million individuals working in Italy, which allows for disentangling voluntary from involuntary job moves, I construct two sector-specific occupational indicators of job security that I attach to each employment spell: one capturing the risk of dismissal and the other conveying a measure of expected tenure. Then, employing an identification strategy that exploits collective dismissals as exogenous variations and the staggered treatment timing at which they occur, I find that displacement leads to an increase in the risk of dismissal inherent to post-displacement occupations of about 2.36 percentage points and a lower expected tenure of around 155 days. These effects correspond to an approximately 12-to-13% decline in job security relative to pre-displacement averages.

JEL codes: J63, J62, J24.

Keywords: Job displacement, Occupational sorting, Job security, Unemployment scarring, Job ladder.

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1 Introduction

The idea that unemployment can have long-lasting repercussions on workers' careers has been around for several decades. Since Heckman and Borjas (1980), who first delineated the methodological challenges involved in the estimation of the impact of job loss on workers' labor market outcomes, an extensive body of literature has emerged to ascertain the presence and evaluate the extent of displacement's effects on earnings (Jacobson et al., 1993), wages (Ruhm, 1991; Arulampalam, 2001), and prospective unemployment (Gregg, 2001). Yet, one aspect that remains underexplored is how job sorting following displacement affects subsequent occupational outcomes, which may ultimately contribute to the recurring unemployment cycles observed in dismissed workers' careers.

This study starts from the intuition that displaced workers' heightened exposure to recurring unemployment may be closely linked to the occupations they find after losing their jobs. In particular, displacement is a disruptive event that may confine workers to a labor market of limited opportunities, where they can only compete for lower-quality occupations. This may occur, for instance, because more attractive vacancies open less frequently and take longer to fill, making them de facto unattainable for recently dismissed workers with poor outside options. If the set of accessible occupations to displaced workers tends to include the generally less desirable ones, then these occupations may not only offer lower compensation but also come with reduced job security. Consequently, sorting into these occupations may, in and of itself, constitute a fundamental driver for the propagation of the adverse effects of displacement.

The proposed analysis tests the hypothesis that displacement leads workers to sort into occupations characterized by lower job security than their pre-displacement occupations. In other words, the objective is to assess whether displacement induces workers to fall along the occupational ladder. To test this hypothesis, I use a dataset containing the employment histories of approximately four million individuals working in Italy. A key feature of this dataset is the availability of information on the reason for each job termination, as well as on the start and end dates of each employment spell, which allows me to disentangle voluntary from involuntary job moves and precisely measure each job tenure. This information was crucial to the development of two job security indicators, which I assign to each sector-specific occupation in the sample (hereafter loosely referred to simply as "occupations"): one capturing the risk of unemployment and the other measuring expected tenure.

Several factors may account for differences in job security across occupations: i) unions may have a stronger presence in some sectors and occupations and may be more effective in protecting tenured workers rather than new hires, making entry-level positions more vulnerable to dismissal; ii) entry-level occupations may also be subject to less stringent employment protection laws; iii) some occupations may be more exposed to the risks of automation or offshoring; iv) occupations in certain sectors, such as manufacturing and hospitality, may be more sensitive to business-cycle fluctuations; v) some sectors, by their own nature, tend to have a higher prevalence of seasonal or temporary work. The job security measures developed for this analysis are designed to capture this heterogeneity across

occupations, which constitutes the ultimate focus of this study.

I estimate the effects of displacement on occupational job security using a difference-in-differences analysis that leverages collective dismissals as exogenous variations and the staggered treatment timing at which they occur to define treatment and control groups. The treatment group then consists of workers who have already experienced a collective dismissal, while the control group includes those who will experience one at a later stage. The motivation for using later-treated individuals, rather than never-treated ones, as a control group rests on the assumption that workers subject to collective dismissals—albeit at different points in time—are more likely to share similar unobservable characteristics, making them more suitable comparison units. Nonetheless, I also conducted a supplementary analysis using a more conventional approach that relies on never-treated individuals as controls. The resulting estimates are used as a benchmark for the main specification. To the best of my knowledge, the proposed approach to the estimation of the costs of job loss is a novelty in the literature. Moreover, using collective dismissals as instruments for unemployment occurrence offers an important advantage over existing designs that rely on mass layoffs as exogenous shocks. While both types of events are firms' workforce readjustments motivated by economic or organizational reasons—and are therefore plausibly orthogonal to dismissed workers' unobservable abilities—collective dismissals are typically smaller-scale shocks than mass layoffs.¹ As a result, concerns about substantial spillover effects in local labor markets, often raised in the context of mass layoffs (e.g., [Cederlöf, 2024](#)), are considerably mitigated.

The main findings indicate that displacement leads to a significant decline in the job security of the subsequent occupations found by displaced workers after dismissal. In particular, these occupations feature a significantly higher inherent risk of unemployment of approximately 2.36 percentage points and a reduced expected tenure of about 155 days. To give a sense of the magnitude of these effects, the increase in unemployment risk corresponds to a 12.8% rise over the average value before displacement, while the drop in expected tenure is approximately equal to a 11.7% decline with respect to pre-displacement averages. Notably, these effects seem to be relatively stable and persistent over time.

Moreover, examining the heterogeneity of these findings across different subgroups, such as gender, level of education, and geographical region of employment (North, Center, and South of Italy), I find estimates that suggest a larger decline in occupational job security for women and for individuals working in the northern regions of Italy. On the other hand, estimates for workers with different levels of educational attainment do not provide a clear-cut picture, although the negative effects of displacement appear more persistent among lower-educated individuals. Nevertheless, wide confidence intervals do not allow for claiming statistically significant gaps among any of these subgroups.

¹ Section 2.1 provides an extensive discussion of collective dismissal procedures. In short, these procedures, within the Italian legislation, apply to companies with more than 15 employees that intend to dismiss at least five employees. This implies, for instance, that if a company of 50 employees wants to dismiss 10% of its workforce (i.e., five employees), this would qualify as a collective dismissal. On the other hand, the literature (see, for example, [Bertheau et al., 2023](#)) usually defines mass layoffs as a 30% reduction in a firm's workforce within a year. Hence, collective dismissal may also entail firm-level shocks that are smaller in size with respect to mass layoffs.

More in detail, examining average post-treatment effects—defined as the mean of the estimated coefficients across all post-treatment periods—I find that men experience a 2.1 pp increase in unemployment risk, while women seem to endure more severe consequences of displacement, with an increase of 2.71 pp. The decline in expected tenure is in line with these results, with men and women facing reductions of approximately 163 and 149 days, respectively. Turning to the dynamics of these effects, the decline in occupational job security remains relatively persistent over time for women, whereas men exhibit a gradual recovery. Differences between higher-educated (those holding at least a college degree) and lower-educated individuals, instead, seem to emerge only in later periods and can only be spotted clearly for the expected tenure indicator. On average, the expected tenure of lower-educated individuals decreases by about 168 days following displacement, compared with a more moderate reduction of approximately 97 days among higher-educated workers. This marked difference is driven almost entirely by the recovery observed among higher-educated individuals in later periods. Finally, workers employed in the North of Italy at the time of dismissal seem to be more adversely affected by displacement than workers in other regions, experiencing an increase in unemployment risk of about 2.53 percentage points and a reduction in expected tenure of approximately 179 days. These effects also appear to be more persistent for workers in the North.

These findings are robust to a comprehensive set of robustness, sensitivity, and placebo checks addressing several potential concerns. First, I assess the accuracy of the constructed indicators, given that sectors and occupations in the sample differ in their level of representation. Second, I examine whether including collective dismissals as involuntary job-loss events when constructing the indicators may raise meaningful endogeneity concerns. Third, I test whether the results are driven by a small number of outlier observations. Fourth, since the baseline analysis focuses on workers aged 30 to 54 (as workers at the beginning or end of their careers might react substantially differently to displacement than prime-age workers), I extend the sample to include all individuals aged 20 to 64. Finally, to verify that a significant variation happens in correspondence with the treatment timing, I arbitrarily reassign the treatment three years prior to its actual occurrence. The first four exercises yield estimates that are close to those from the main specification, while the placebo test confirms that the estimated effects reflect a fundamental change occurring in correspondence with the displacement year.

This study contributes to the literature on the costs of job loss by estimating the causal effects of displacement on the job security of the occupations into which dismissed workers sort after displacement. While previous research has thoroughly documented the negative consequences of unemployment on workers' career trajectories (Ruhm, 1991; Jacobson et al., 1993; Stevens, 1997; Arulampalam et al., 2000; Arulampalam, 2001; Gregg, 2001; Burgess et al., 2003; Gregg et al., 2004; Gangl, 2006; Eliason and Storrie, 2006; Raaum and Røed, 2006; Mroz and Savage, 2006; Von Wachter and Bender, 2006; Oreopoulos et al., 2012), this analysis shifts the focus to the role of the post-displacement search and matching process as a mechanism shaping these outcomes. The central idea advanced in this study is that displacement may constrain workers' options, bounding them to accept lower-quality

occupations with intrinsically lower job security. Several explanations may account for these effects unfolding. First, human capital models predict that periods of unemployment may erode workers' human capital, thereby reducing the marketability of their skills to potential employers (Mincer and Ofek, 1982; Pissarides, 1992). Second, unemployment may carry a stigma (Vishwanath, 1989; Gibbons and Katz, 1991; Omori, 1997), which translates into a negative signal of labor productivity during the screening process. Third, displacement may untie high-quality worker–firm matches, especially when workers' abilities and employers' skill requirements are well aligned. Fourth, job ladder models posit that workers try to “climb” towards higher-paying and more secure jobs over their careers (Burdett and Mortensen, 1998), but displacement may destroy their progress, pushing them into less-secure and lower-paid jobs (Pinheiro and Visschers, 2015; Jarosch, 2023).² Being dismissed thus entails falling to the bottom of the job ladder, where rungs are more slippery and the probability of experiencing repeated unemployment spells is higher.

While all these mechanisms may be at play simultaneously, the underlying idea of this study is closer in spirit to the latter one: sorting into occupations with inherently lower job security increases exposure to future displacement risk, thereby giving rise to a job-ladder-type chain of events. Building on prior theoretical work, I empirically quantify the decline in occupational job security leveraging a novel research design that exploits the staggered timing of collective dismissals as exogenous shocks and detailed information on employment tenure and reason for termination to construct occupational-level job security measures.

Concluding, it is worth noting that investigating heterogeneity in occupational job security is particularly informative because, as shown by Pinheiro and Visschers (2015), unemployment scarring can arise even when heterogeneity is assumed to be present solely on the demand side (imposing homogeneity in workers' inner characteristics). The evidence presented here corroborates this idea, which, from a theoretical perspective, demonstrates that the labor demand alone can, in principle, explain the persistent adverse consequences observed in displaced workers' careers. Similarly to this paper, Figueiredo et al. (2026) measures job security by examining whether displaced workers find permanent or temporary jobs afterward. This study, instead, classifies each occupation according to its inherent job security, allowing for richer heterogeneity. Within this framework, occupations serve as a fundamental channel through which displaced workers are differentially exposed to the risk of recurrent unemployment.

This paper is structured as follows: Section 2 describes the data, Section 3 outlines the two measures of occupational job security, Section 4 motivates the empirical strategy, the results are presented in Section 5, followed by robustness checks and placebo in Section 6, and Section 7 concludes.

² The positive correlation between job security and wages has been widely discussed in the literature on the lack of compensating differentials (Mayo and Murray, 1991; Winter-Ebmer, 2001; Bonhomme and Jolivet, 2009). For example, Mayo and Murray (1991) shows that workers sort into large or small firms based on their unobservable quality. In this context, smaller firms, which typically offer both lower wages and less stable employment, may attract workers with generally more unstable labor-market histories, who are willing to accept worse employment conditions given their poor outside options. Unlike this work, I do not measure job security at the firm level but rather at the sectoral-occupational level. However, Haltiwanger et al. (2022) shows that most of the wage dispersion (and thus most of the job security dispersion) occurs at the sectoral level, which supports the relevance of the indicators (at the sectoral and occupational level) used in this study.

2 Data

This study relies on a sample extracted from the “Comunicazioni Obbligatorie” database,³ which records mandatory notifications that Italian employers (or their intermediaries) are required to submit to the competent regional or national authority regarding important workforce variations, such as hirings, firings, or contract conversions.⁴ The sample accounts for around 22 million observations, which are obtained from a random selection of 4 million individuals employed within the Italian labor market, both in the private and public sectors, from 2010 to 2022.⁵ The data contains the complete employment histories of the selected individuals throughout the considered time frame. In its original structure, each row reports information on one single employment spell, which includes the starting and ending dates,⁶ the contract type (temporary or permanent, full-time or part-time), the occupation, the sector of employment, and notably, the reason for job termination, which is an essential element for the computation of the outcome variables. The dataset also includes several time-invariant individual characteristics such as gender, year and region of birth, and education.⁷

To prepare it for the analysis, I restructured the dataset into a yearly panel format, where every year reports each individual’s main employment.⁸ Next, I extracted the sample of interest, which consists of individuals who experienced only one collective dismissal during the considered period. I limited the analysis to workers with a single collective dismissal to avoid complications in the interpretation of the estimates that may arise in the presence of multiple treatments.⁹ Additionally, I excluded workers who started a new job before their prior contract’s official termination with the dismissing firm, those who were dismissed from a job that was not identified as the main one, as well as those who returned to the same firm that had previously dismissed them. This last cut was done to avoid the possibility of leaving in the sample cases of workers’ reinstatements when the collective dismissal is later ruled illegitimate by a court. To further homogenize the sample, I restricted it to individuals aged at least 30 in 2010 and not older than 54 in 2022. This choice was motivated by the fact that young and senior workers may respond differently to displacement with respect to prime-age workers: younger individuals, early in their careers, frequently move across many jobs regardless of

3 More specifically, the name of the sample is “Campione Integrato delle Comunicazioni Obbligatorie,” in short, CICO.

4 Note that individuals appear in the dataset only if at least one of these events occurs within the sampled period.

5 The random selection is performed by extracting every individual born on the 1st, the 9th, the 10th, and the 11th day of any month and any year.

6 Note that the hiring date may be antecedent to 2010, the first year observed in the dataset. The expected tenure indicator is computed based on this information. This implies that employment spells are only right-truncated, so censoring arises only along this dimension. Furthermore, assuming censoring is identically and independently distributed across occupations, this should not represent a major concern.

7 The reported level of education in the dataset refers to the one at the time of the last registered employment spell. This is a non-trivial caveat, as it is not possible to know whether individuals upgraded their education once they started working.

8 Similarly to Card et al. (2013), I restricted the sample to keep a single dominant record per year. However, as earnings are not the focus of this study, I defined the main job as the one with the longest employment spell within each year. To identify main jobs, I sequentially retained the employment record that displayed: 1. the highest number of employment days within the year; 2. the longest employment spell overall (spanning throughout the whole sampled period); 3. full-time jobs were preferred to part-time ones; 4. highest income (however, income is reported only for employment spells that started after 2010). When two or more employment spells were found to be equal in every one of these categories, I picked one at random.

9 However, including individuals who were treated more than once does not sensibly change the results, see Figure B1. This may be due to the relatively low number of individuals who experienced a collective dismissal more than once in the sampled period (only 1.7% of the total).

dismissal, while older workers may instead react to displacement by anticipating their retirement.¹⁰ Finally, to extract a balanced dataset, I also removed individuals who had gap years in their reported employment history (i.e., a full calendar year without a single employment day), which implies eliminating the long-term unemployed from the sample. The main rationale for excluding these types of workers is that individuals who remain unemployed for an extended period of time may carry unobservable characteristics that deeply differentiate them from workers with more stable employment histories, possibly introducing major confounders into the analysis.¹¹ After making all these adjustments, and after excluding those individuals who experienced a collective dismissal within the first three years of the studied period (to allow for at least three pre-treatment years), the final sample accounts for 8,256 treated individuals observed over 13 years, which corresponds to a total of 107,328 observations.¹² Summary statistics pertaining to this sample are shown in Appendix A from Table A1 through Table A5. Most of the workers in the sample are employed in northern Italy ($\approx 58.4\%$ of the total), have a high school diploma ($\approx 52\%$), are male ($\approx 67\%$), and, on average, are about 42 years old. The size of all treatment cohorts is sufficiently large, although earlier cohorts, from 2013 to 2017, are considerably larger than later ones. The distribution of individuals across treatment cohorts is reported in Table A6 in Appendix A. The main specification will only make use of this group of units, exploiting later-treated units to build control groups that vary by cohort and calendar time.

Nevertheless, the analysis can also be conducted by employing a group of never-treated individuals as control units. In this case, untreated units must be selected based on observable characteristics that make them suitable comparison units for extrapolating untreated potential outcomes. Since the estimator used in the analysis, proposed by [Callaway and Sant’Anna \(2021\)](#), already incorporates a propensity score matching procedure, it is not necessary to identify suitable control units prior to the estimation stage. When never-treated units are included, the resulting sample comprises 3,364,361 observations and 258,797 individuals, of whom 7,449 are treated.

In the following sections of this paper, I will mainly focus on estimates obtained with the sample that includes treated units only. The motivation for this choice will be outlined in Section 4. Nonetheless, as identification strategies that employ never-treated units have been the standard, the charts and the tables showing the results pertaining to the specifications that make use of the untreated

10 For example, [Farber et al. \(2019\)](#), who studies the role of employment and unemployment histories in callbacks to job applications, find that younger and older applicants have a lower callback rate than prime-aged applicants. Furthermore, also in [Bertheau et al. \(2023\)](#), only workers who are at most 50 years old in the year preceding displacement are retained to limit the influence of early retirement programs. Differently from this approach, however, I preferred to further homogenize the age composition of the sample, given that later-treated units are used in this work to construct counterfactual potential outcomes and thus might generally be slightly older than already-treated units. In [Bertheau et al. \(2023\)](#), instead, the authors employ a more standard identification strategy that exploits never-treated units as controls.

11 In addition, other papers in the literature, such as [Ruhm \(1991\)](#), have followed this approach of considering only “individuals with fairly strong attachments to the labor force”. Furthermore, [Eriksson and Rooth \(2014\)](#), for example, finds that potential employers attach a stigma to contemporary unemployment spells that last at least nine months, which negatively affects their hiring decisions of long-term unemployed individuals.

12 As opposed to what is common practice in the literature on the cost of displacement (e.g., see [Jacobson et al., 1993](#)), the sample was not restricted to include only individuals with at least three years of tenure in the firing firm at the time of dismissal. This choice was motivated by the fact that outcome variables, described in Section 3, are time-invariant indicators of job security and, thus, only vary when workers change employment. Hence, keeping only individuals who have not changed firm over the three years preceding the collective dismissal would have implied ending up with a complete absence of variation in the outcome variable for those periods, which would have made these pre-treatment coefficients impossible to compute.

units' sample will be contextually reported in Appendix B, serving the purpose of being a benchmark for the main results.

2.1 The Italian institutional setting of collective dismissals

The collective dismissal (*licenziamento collettivo*) procedure in Italy is the process through which a company can dismiss a substantial number of employees because of economic, organizational, or structural reasons. In the dataset that I employ for this analysis, 0.62% of all job separations are filed as collective dismissals.

Collective dismissal procedures are regulated by the Law 223/1991, which was further refined by the legislative decree n.151 of the 26th of May 1997, with the intention to homogenize the legislation on the subject across the European Union's member countries. These procedures apply to companies with more than 15 employees who intend to dismiss at least five employees within 120 days and involve several steps.¹³ First, the company must send a written notification to the trade unions and the relevant labor authorities, providing detailed information about the reasons for the dismissals, the number and roles of employees affected, and the previewed timeline. Then, within seven days, trade unions may propose to re-examine the reasons for the collective dismissal and discuss possible alternative solutions with the company, such as temporary layoffs, retraining, or reduced working hours. This consultation phase lasts a maximum of 45 days (or about 22 days in the case the procedure involves less than ten employees). If no agreement is reached by the deadline, administrative authorities may initiate a second consultation phase, which must end within 30 days (15 in case the procedure involves less than ten employees). Once both consultation phases are over, the company can proceed to dismiss the workers that it deems to be redundant. To do so, the company must apply criteria that had been previously determined with union agreements. Typically, these criteria include the length of service (newcomers shall leave the company first, similar to the "last-in-first-out" rule present in the labor market legislation of other European countries), the presence of family responsibilities, and the company's operational needs. Finally, dismissed employees are entitled to severance pay and may qualify for unemployment benefits. Furthermore, the law provides that firms that hire workers who were previously displaced in the course of a collective dismissal are entitled to substantial tax breaks.

One important caveat is that, while younger workers may have fewer family responsibilities and shorter tenure, which are factors that might increase their likelihood of being subject to collective dismissals, the "pensionability" criterion is among the most commonly used in such procedures. This consideration partly explains why I chose to exclude both younger and older workers from the analyzed sample, as discussed in the data section.

¹³ One potential concern is the possible presence of anticipation effects. If workers are notified of their imminent dismissal, they may begin searching for a new job before the actual displacement date. To address this issue, I examine whether including workers who start their subsequent job within a window of 120 days before their actual dismissal affects the results. While effects are slightly diminished, including these workers does not change the estimates substantively. The corresponding results are reported in Figure B2 in Appendix B.

3 Measuring occupations' job security

Taking advantage of the richness of the original dataset, I developed two indicators to gauge the level of job security associated with each sector-specific occupation¹⁴ in the sample: one assessing the risk of unemployment and the other measuring expected tenure. These indicators were calculated for 2131 sectors and 531 occupations separately, and then averaged to derive a single measure of job security for each sector-specific occupation (hereafter loosely referred to as "occupations" for the sake of simplicity). This process yields an indicator of job security for 9316 distinct occupations present in the final dataset of treated units.

To construct the two job security measures, I followed slightly different procedures. For the unemployment risk indicator, I exploited a crucial piece of information in the dataset, which is the reason behind each job termination. Then, observing the annual survival rates of each individual job, I categorized termination events, assigning a value of one to involuntary separations and a value of zero to voluntary job transitions and ongoing employment spells.¹⁵ Subsequently, I computed the share of involuntary separations for each occupation and sector separately, which I then averaged to finally get a time-invariant measure that captures the risk of unemployment attached to any given sector-specific occupation. This process is summarized by the formula below:

$$y_{jk} = \frac{\left(\frac{\sum L_{itj}}{N_j} + \frac{\sum L_{itk}}{N_k} \right)}{2}$$

where y is the unemployment risk indicator pertaining to each occupation, defined as the combination of sector j and occupation k of individual i , L is a dummy variable that takes the value zero for voluntary job moves and ongoing work spells within the considered year t or the value of one when the worker is subject to a layoff, and N is the number of observations for any given sector and occupation.

I developed the second indicator, conveying a measure of expected tenure, in a similar fashion. However, contrary to the previous case, constructing this indicator did not require an inspection of jobs' annual survival rates, but it was sufficient to calculate their length, knowing the hiring and separation dates provided in the data. Nevertheless, information on the separation reasons was still useful, as I needed to filter out from the computation employment spells that ended with voluntary job moves.¹⁶ Once I obtained these mean values pertaining to each occupation and sector separately, I averaged these employment durations to get a single measure for the expected tenure attached to

¹⁴ Sectors refer to 6-digit ATECO codes, while occupations are defined as "the set of activities that an individual must perform in carrying out their job" (as defined by the Italian National Institute of Statistics, according to the classification CP2011). These activities imply knowledge and skills and are codified and regulated.

¹⁵ To conduct this operation, employment spells that reported a termination reason that left unclear the categorization into voluntary or involuntary separations, such as firm closures, were discarded. In the case of firm closures, for example, it was not possible to determine whether the firm closure entailed an actual shutdown of the firm, with all its employees being made redundant, or whether the enterprise was acquired by another company resulting, therefore, only in a transfer of employment from one employer to another. Eliminating such cases discards about 4% of all the employment spells in the sample utilized for the computation of the job security indicators.

¹⁶ Employment spells with unclear separation reasons, as mentioned before, were also excluded.

each sector-specific occupation, as illustrated before for the other indicator. The formula used to compute this second measure, in fact, closely resembles the previous one:

$$y_{jk} = \frac{\left(\frac{\sum d_{ij}}{N_j} + \frac{\sum d_{ik}}{N_k} \right)}{2}$$

where y is now the expected tenure indicator for each occupation, while d denotes the average number of days individuals spent working in that occupation.

Both indicators should reflect the heterogeneity in job security associated with being employed in different occupations.¹⁷

Finally, it is important to remark that both measures of job security are time-invariant and thus convey the level of job security for each occupation for the whole considered time span. The rationale for this choice is to avoid mechanically generating a dip in occupational job security at the time of dismissal, followed by a subsequent increase, which would inflate the measured risk of dismissal in the pre-treatment period and deflate it in the post-treatment period. Moreover, allowing the occupational job security indicator to vary over time would make it procyclical, thereby downplaying the risk of dismissal associated with each occupation during economic expansions and augmenting it during recessions. Since the objective of this paper is to estimate the fall along the occupational ladder that follows displacement, workers who do not switch occupations should face the same level of occupational risk as they would in the absence of displacement.

Table 1 shows mean values of both outcome variables before and after treatment, while Figure 1 displays the distribution of the two outcome variables' values in the final sample, regardless of the treatment period.¹⁸

Table 1: Mean value of the outcome variables before and after the treatment's occurrence

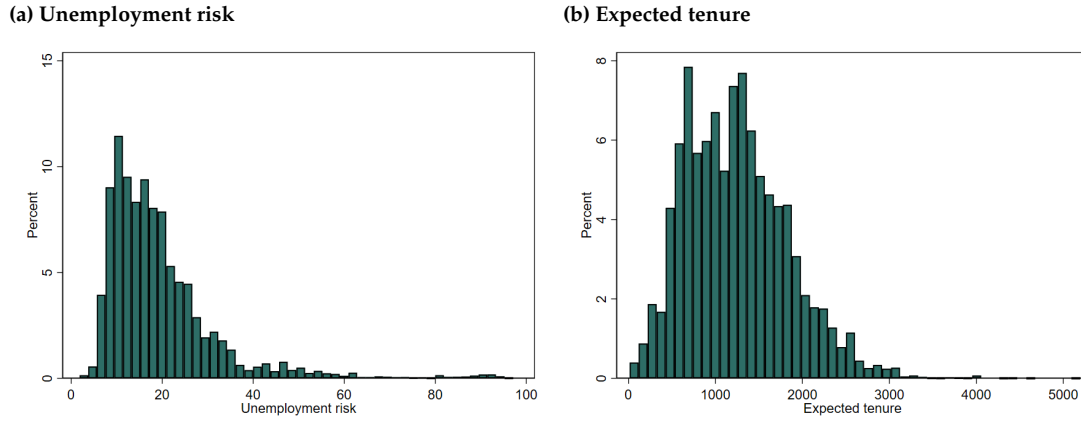
	Unemployment risk	Expected tenure
Before displacement	18.43 (11.19)	1330.47 (600.18)
After displacement	20.46 (13.53)	1166.369 (564.69)

Notes: Standard errors in parentheses. The table displays the average value for the two outcome variables (unemployment risk and expected tenure) before and after displacement takes place. The numbers refer to the 8256 individuals in the final sample of treated units.

¹⁷ To facilitate the comprehension of these measures, some examples of occupations characterized by high job security are professionals who work in the banking sector and engineers/designers who work in the automotive or aircraft production sector, whereas examples of low job security occupations are day laborers who work in the farming sector and professionals who work in the cinematographic industry or theaters.

¹⁸ See Figure B3 in Appendix B to compare with pre and post-treatment period figures.

Figure 1: Distribution of the outcome variables' values in the sample of treated units



Notes: The two charts show the distribution of the two outcome variables' values in the final sample of treated units. The charts, therefore, display values belonging to both pre- and post-treatment periods. The figures refer to the 8256 individuals in the final sample of treated units.

3.1 Validation of occupational job security measures

To evaluate whether the proposed job security measures reasonably reflect the likelihood of dismissal intrinsic to each occupation, I categorized the sample of dismissed workers (as a result of a collective dismissal) into two groups: those who sorted into a new occupation with a higher job security than their pre-displacement one, and those who found an occupation with a lower or equal job security. Then, I computed workers' unemployment days for each year following the first one in which they found their new jobs and estimated the following regression:

$$y_{it} = \beta_1 D_i + \beta_2 X_i + \delta_t + \varepsilon_{it} \quad (1)$$

where y are worker i 's nonemployment days in year t , D is a dummy variable that takes value 1 if the worker found a worse occupation in terms of job security following the collective dismissal, and zero otherwise, X is a vector of workers' time-invariant characteristics (age, gender, level of education), and δ are year dummies.

Table 2 shows the results of this exercise, indicating that workers who found an occupation featuring a worse unemployment risk (*expected tenure*) with respect to the pre-displacement occupation suffer, on average, 4.5 (3.1) more unemployment days with respect to those who sort into a more secure occupation. Compared to the average worker who moves to a more secure occupation, this corresponds to approximately 31% (21%) more nonemployment days. I interpret this result as descriptive evidence that the proposed indicators successfully provide a reasonably accurate measure of the intrinsic probability of involuntary dismissal of each occupation in the sample.

Table 2: Difference in days of nonemployment among workers who found a lower job security occupation after dismissal vs workers who found a higher job security occupation

Variables	Nonemployment days	
	Unemployment risk	Expected tenure
Worse outcome after treat	4.46*** (0.50)	3.12*** (0.50)
Time-invariant observables	✓	✓
Year dummies	✓	✓
Constant	24.06*** (3.65)	24.46*** (3.66)
Observations	50,561	50,561
R-squared	0.0145	0.0137
***p<0.01, **p<0.05, *p<0.1		

4 Identification and Empirical Strategy

The aim of this analysis is to assess the impact of displacement on occupational job security. This task poses substantial methodological challenges, as dismissals may be correlated with unobservable workers' characteristics that may influence their future job prospects. Partialling out these features is therefore fundamental to get causal estimates of the effects of displacement on workers' labor market outcomes. To tackle this issue, researchers have typically resorted to firm-level exogenous shocks, such as mass layoffs, as instruments for unemployment occurrence. The rationale is that these shocks are considered exogenous to workers' characteristics as, in these contexts, firms do not select the pool of workers to be dismissed based on their performance but rather according to economic and organizational needs. Hence, in the context of mass layoffs, all types of workers may be exposed to displacement risk, irrespective of their individual attributes.¹⁹ However, [Cederlöf \(2024\)](#) questions the validity of this approach, arguing that mass layoffs may generate significant spillover effects at the local, sectoral, or industrial level, which can thereby exacerbate the negative effects of displacement. This concern is clearly more relevant when mass layoff represents a sizeable portion of the local labor market. Studies that employ mass layoffs as sources of exogenous variation typically define them as a reduction of at least 30% of a firm's workforce within a given year. Nevertheless, these works usually rely on data lacking information on the reason for employment termination, which requires researchers to rely on large workforce downsizes to correctly discriminate between job separations due to economic reasons and individual layoffs.²⁰ This study, instead, uses a dataset that contains this information, allowing to disentangle between these two separation reasons. This feature constitutes a significant advantage relative to previous research. Furthermore, collective dismissals, while

¹⁹ One of the earliest published articles in the literature adopting this approach was [Jacobson et al. \(1993\)](#), which was followed by a long list of works that used the same, or somewhat similar, approaches.

²⁰ This is, for instance, explicitly outlined in [Bertheau et al. \(2023\)](#), where to correctly specify the treatment group they must impose a 30% annual drop in a given firm's workforce in order to avoid mischaracterizing voluntary separations as layoffs.

preserving the nature of being firm-level shocks motivated by economic or organizational reasons, are generally smaller in scale than mass layoffs, which alleviates concerns about spillover effects being too pronounced in local labor markets.

The following analysis employs a difference-in-differences event study design in which the control group consists of workers who have not yet been displaced through a collective dismissal. The motivation for choosing not-yet-treated units as a control group lies in the idea that workers who undergo a collective dismissal, even at different points in time, are more likely to share more similar unobservable traits with each other than with those who were never treated. Consequently, estimates obtained using not-yet-treated units to derive counterfactual potential outcomes can be presumed to be less vulnerable to biases stemming from unobservable time-varying factors, as these types of confounders are more likely to be implicitly netted out. In the next section, I will therefore only present results obtained from regressions that exploit not-yet-treated units as a comparison group, but to benchmark these estimates, I contextually carried out the analysis using never-treated individuals as control units. These results will be reported in Appendix B.

The chosen methodology to retrieve the event-study estimates is the estimator proposed by [Callaway and Sant'Anna \(2021\)](#) (henceforth CS). This methodology is particularly well suited for this empirical framework for a couple of reasons. First, as mentioned above, it is robust to heterogeneous treatment effects, which is crucial in this setting given that workers in the sample are employed in very diverse occupations and face dismissals at different points in time, therefore encountering varying contingent labor market conditions and regulations. Second, this estimator maximizes the use of not-yet-treated units as part of the control group. This is a distinctive advantage over alternative estimators designed to deal with heterogeneous treatment effects that have been devised in recent years. For instance, [Sun and Abraham \(2021\)](#) only exploits the last-treated cohort as a control group, while the CS estimator, leveraging the "no anticipation" assumption, employs every cohort that is yet to be treated. This feature of the CS estimator is particularly valuable given that the dataset includes fewer treated units in later periods. It needs to be remarked that the CS estimator employs covariates to generate a propensity score that matches units based on a defined set of observable characteristics fixed at their pre-treatment year values. The parallel trends assumption required for identification is then the following: conditional on a set of observable characteristics, treated and control units followed similar occupational job security paths prior to dismissal; in the counterfactual scenario where displacement would not have occurred, these trajectories would have remained unchanged. Given this conceptual framework, the main specification is:

$$y_{itg} = \sum_{e, e \neq -1} \beta_e \times D_{it} \times \mathbb{1}(t = g + e) + \pi X_i + \varepsilon_{itg} \quad (2)$$

where y is the unemployment risk indicator or the expected tenure in a given occupation, faced by individual i of cohort g (the cohort is defined by the treatment year) in year $t \in \{2010, \dots, 2022\}$, D is a dummy equal to 1 for the years after the collective dismissal and equal to zero before, e is event

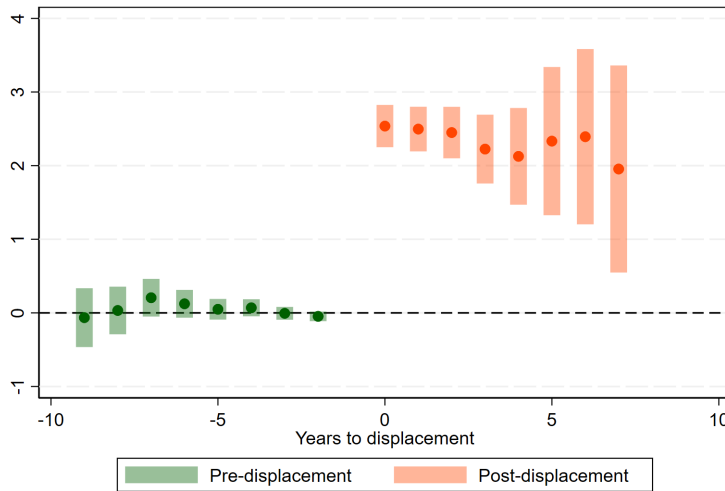
time (with $e = 0$ corresponding to the year of treatment), X is a vector of pre-treatment ($e = -1$) time-invariant covariates (including full-time and temporary employment status, macro-region of work, age, gender and education).²¹ Standard errors are clustered at the worker level, and the chosen estimation method is the doubly robust difference-in-differences estimator based on stabilized inverse probability weighting and ordinary least squares as in [Sant'Anna and Zhao \(2020\)](#). The parameter of interest is β , and it can be interpreted as the effect of displacement on the subsequent job security (either in terms of unemployment risk or expected tenure) attached to the occupations found by dismissed workers following dismissal.

Furthermore, to benchmark the results, I performed the estimation (although using an unconditional specification alongside cohort-worker and cohort-year fixed effects) employing a stacking-by-event methodology as in [Deshpande and Li \(2019\)](#) or [Cengiz et al. \(2019\)](#).

5 Results

Figure 2 presents the results obtained from estimating equation (2) for the first of the two outcome variables, the unemployment risk. The chart displays the expected change in the unemployment risk linked to the occupations held by treated units compared to the variation in the control group before and after the treatment. In other words, post-treatment estimates reflect the average treatment effect of displacement on the dismissed workers' expected unemployment risk associated with the occupations that they find after being dismissed.

Figure 2: Unemployment risk



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

²¹ Notice that for the analysis that exploits never-treated units as controls, I fixed covariates to the last year before any unit in the sample had been treated, which is 2012.

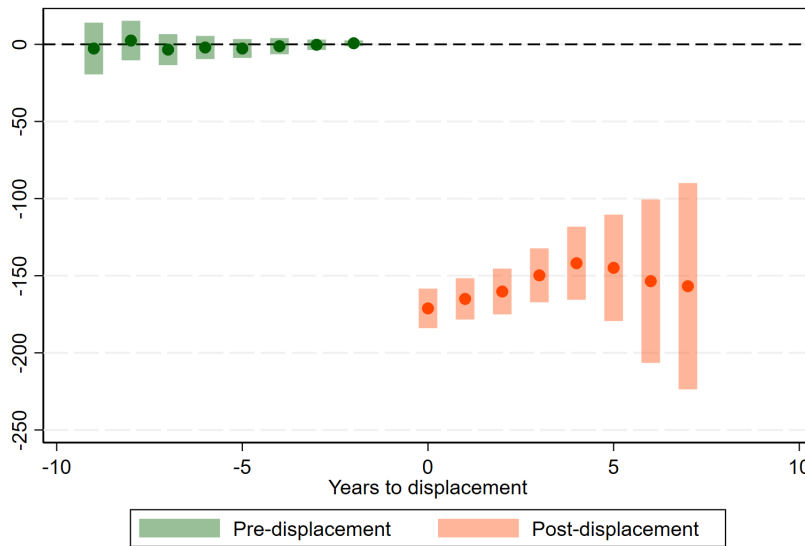
All pre-treatment coefficients are close to zero and not significant, which suggests that the parallel trend assumption holds. Post-treatment estimates are, instead, all positive, significant, and relatively stable over time, ranging between 2.56 pp in the first period after the treatment and 2.08 pp in the last period. Note that the chart omits the value corresponding to $e = 8$, which is also excluded from all regression results presented next. The reason for not showing this estimate is that its value is obtained by exploiting only one treated cohort (that of individuals treated in 2013) and one not-yet-treated cohort as a control group (that of those treated in 2022). However, as reported in Table A6 in Appendix A, the 2013 cohort accounts for 1165 individuals, while the cohort of 2022 counts only 236 individuals. This implies that there is about one control unit for every five treated ones. Given the limited size of the last-treated cohort and the relative imprecision in the computation of this estimate, I preferred to leave it out from the presented results. An analogous motivation applies to two pre-treatment coefficients that have also been discarded from the chart, which are those corresponding to the periods $e = -10$ and $e = -11$.

Estimates obtained using never-treated individuals as controls (shown in Figure B4 of Appendix B) are similar to those retrieved with not-yet-treated units, although coefficients are slightly smaller in magnitude and display a mild (but noticeable) recovery over time. Additionally, a further distinction between the results derived from the two different specifications is that, while the average of the pre-treatment estimates is still close to zero (0.125), it is instead jointly positive and significant for the never-treated case (even though this result is mainly driven by the coefficients that are farther away from the treatment period). Taking stock of these findings, while using never-treated units as controls yields post-treatment estimates that are comparable to those obtained with not-yet-treated units, an examination of pre-trends suggests that, as hypothesized in the previous section, the latter type of control units is more appropriate for the empirical setting under consideration.

The interpretation of the estimates shown in Figure 3 is identical to the one of those portrayed in Figure 2. However, this time, the outcome variable of interest is the expected tenure. Again, all pre-treatment coefficients are not significant and close to zero, suggesting that the parallel trends assumption holds. Post-treatment estimates are, instead, negative, significant, and relatively steady over time, ranging from a maximum of 171 days of reduced expected tenure in the year right after the dismissal and a minimum of 142 days five years from the dismissal.

Estimates obtained using never-treated units as a control group are shown in Figure B5 in Appendix B. Similarly to what I observed for the estimates obtained for the unemployment risk indicator, the magnitude of these coefficients is generally inferior with respect to estimates shown in Figure 3. In addition, as before, pre-treatment coefficients are jointly significant but relatively close to zero and have the same sign as post-treatment coefficients. Combined with the results for the unemployment risk indicator, I interpret these findings as providing further evidence that not-yet-treated units constitute more suitable counterfactuals to extract potential outcomes in this empirical framework.

Figure 3: Expected tenure



Note: The chart displays estimates of the average treatment effect of displacement on the expected tenure associated with the main occupations in which “treated” individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Moreover, the analysis has been replicated by means of a stacking by event specification. As mentioned in the previous section, this specification does not feature covariates. Yet, the findings are in line with those from the main specification and are reported in Table A7 of Appendix A. These results can also be compared with those derived using unconditional specifications through the CS methodology. These estimates are plotted in Figures B6 and B7 of Appendix B. The main takeaway from these latter results is that adding covariates with the aim of retrieving credible conditional parallel trends seems to only matter in the case of the expected tenure indicator that uses never-treated units as controls.

Overall, all specifications unequivocally suggest that displacement causes a substantial decline in occupational job security. To give a sense of the magnitude, the increase in unemployment risk corresponds to a 12.8% rise relative to the pre-displacement average, while the reduction in expected tenure represents approximately a 11.7% drop with respect to pre-displacement average. Both indicators therefore imply comparable reductions in occupational job security.

On a final note, before presenting heterogeneous results for different subgroups of workers, two other dimensions of heterogeneity are worth mentioning. First, cohort-specific estimates reveal that the impact of dismissal substantially varies across different cohorts. The 2019 cohort appears to be the most adversely affected, with post-treatment averages of 4.22 pp for the unemployment risk indicator and a reduction of 254 days in expected tenure. In contrast, the 2017 cohort is the least affected, with post-treatment averages of 1.12 pp and a reduction of 119 days, respectively.²² A second dimension

²² The 2016 cohort reports a higher value for the unemployment risk indicator (1.83pp), but a lower one for the expected tenure indicator (116 days). These results are not included in the Appendix and are available on request.

of heterogeneity stems from partitioning the sample into workers who transitioned to their next main occupation within six months since the dismissal and those who took a longer period of time to do so. Notice that, as outlined in the data section, the dataset is constructed to retain only a single dominant employment record per worker for each year. Consequently, it is not easy to precisely calculate the number of unemployment days between the date of the collective dismissal and the employment that immediately follows. Therefore, for the sake of simplicity, here I consider the number of days between two consecutive main employment spells as a proxy for the time it takes a displaced worker to secure a new, relatively stable position. Bearing this caveat in mind, given that human capital depreciates over time and employers often attach a stigma to long-term unemployment, it is reasonable to expect that individuals who take longer to find a suitable job also experience worse outcomes, also in terms of occupational job security. This is indeed the case: I find that those who find their next main job within six months endure substantially weaker effects relative to the baseline, whereas the reverse holds for workers who take more than six months to find their next main occupation. These results are presented in Figure B8 in Appendix B.

5.1 Heterogeneity of results

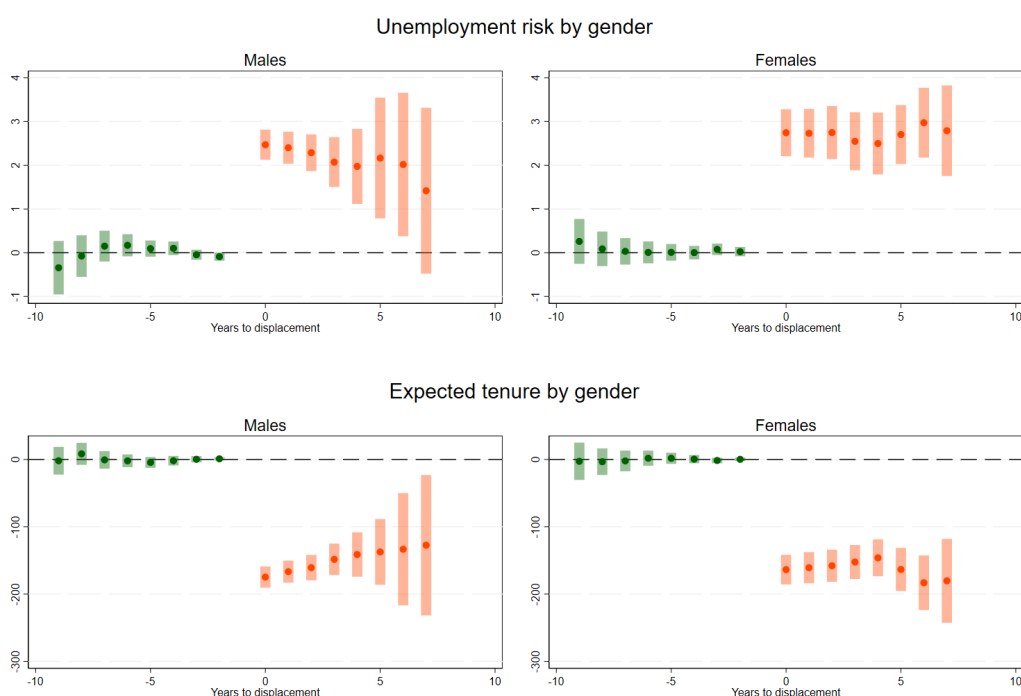
The findings in the previous section document a significant and persistent impact of displacement on occupational job security. A natural follow-up question is to ask whether these adverse effects are homogeneous across various subgroups of workers. Therefore, in what follows, I will examine the impact of job loss on different genders, educational levels, and region of work,²³ to ascertain the presence of possible heterogeneous treatment effects.

First, is there a gender differential in the impact of job loss on the job security of subsequent occupations? Based on the evidence disseminated in the literature, the answer to this question is not straightforward. In fact, on one hand, some studies have shown that women tend to sort into jobs with lower wage premiums (Card et al., 2016) but also a reduced risk of dismissal (Wilkins and Wooden, 2013; see also Table A5). On the other hand, some recent analyses have revealed that job loss leads to more substantial and persistent reductions in earnings for women (Illing et al., 2024). Putting together these findings, two hypotheses could explain the observed gender gap in the cost of job loss on earnings. First, compared to their male counterparts, women after dismissal may end up accepting lower-paid jobs with limited wage growth potential (e.g., part-time jobs). This hypothesis would be consistent with a scenario in which the observed gender gap in earnings following job loss originates primarily from wages or hours worked. Second, displacement may erode women's relative advantage over men in terms of job security. In this latter case, the gender gap in earnings due to displacement would come from a larger decline in job security, reflecting effects on prospective employment stability. Naturally, while both mechanisms could be in place at the same time, the results presented next test only the presence of the latter of these two mechanisms.

²³ For the regressions based on the region of work, workers were classified according to the macro-region (namely, South, Center, and North of Italy) where they were employed in the year prior to the treatment.

Figure 4 presents estimates²⁴ for males on the left and females on the right. The chart in the top right part of the panel illustrates that females experience an average increase in unemployment risk of approximately 2.72 pp, while males (on the top left corner of the chart), instead, face a comparatively smaller average increase of about 2.1 pp. Declines in expected tenure align with these results, amounting to about 163 days for females and 149 days for males. These averages, however, mask rich dynamics in post-treatment effects. In the immediate aftermath of displacement, the drop in occupational job security seems to affect men and women to a similar extent. Over time, however, these negative effects become moderately larger for women, while the opposite pattern holds for men.

Figure 4: Heterogeneous effects of displacement on job security by gender



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk associated with the main occupations in which “treated” individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Estimates derived using never-treated units (shown in Figure B9 in Appendix B) are quantitatively similar to those presented in Figure 4. Both specifications do convey a very similar takeaway message: while the negative impact of dismissal on both outcomes is initially almost equivalent in magnitude for men and women, effects gradually attenuate over time for males while, instead, slightly intensify

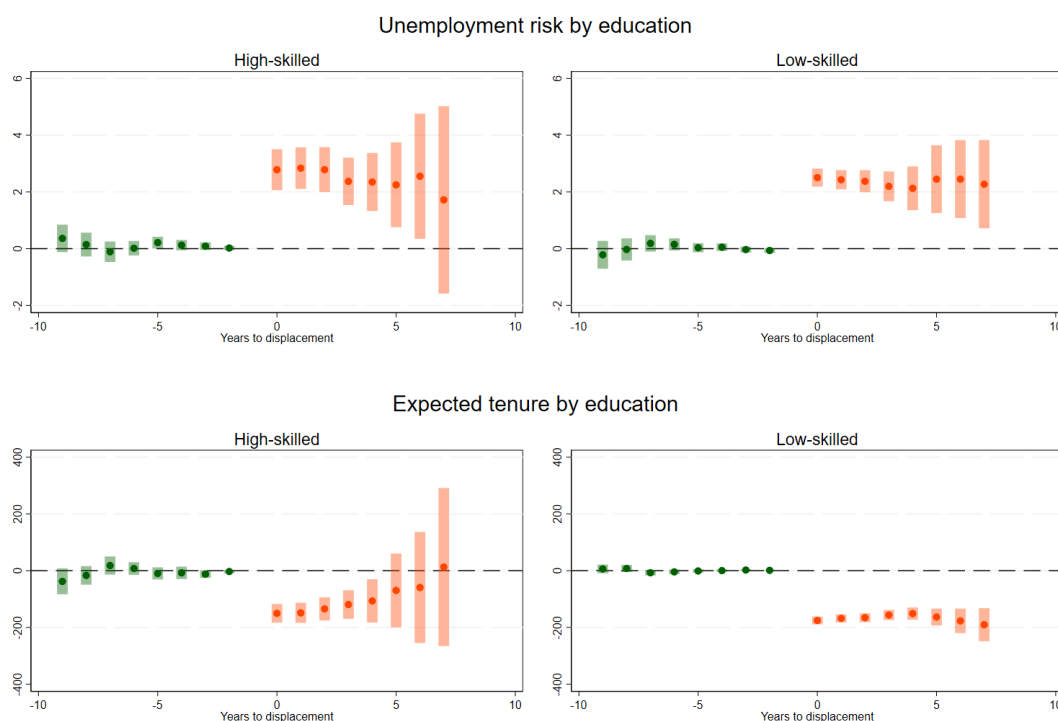
²⁴ These estimates, like those shown in Figures 5 and 6, are derived without including the educational level as a covariate. This decision was made because, when splitting the sample to estimate the model on separate subgroups, the sample size of certain cohorts was not sufficiently large to accommodate all the covariates in the original model. Nevertheless, excluding education from the model does not seem to alter substantially the coefficient of interest.

for females.²⁵

It is important to emphasize, nonetheless, that these considerations solely apply when point estimates are taken into account in isolation. In fact, confidence intervals are too wide to establish any statistically significant differences between men and women. Altogether, these results, therefore, do not provide sufficient evidence to support the hypothesis that women, after dismissal, end up losing their edge over men in terms of occupational job security, even though point estimates do indicate that this might be happening to some moderate degree.

Next, I examine the presence of heterogeneous effects by educational level. To do so, I categorized individuals as high-skilled if they obtained at least a college degree and low-skilled otherwise. The results are displayed in Figure 5.

Figure 5: Heterogeneous effects of displacement on job security by educational level



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk associated with the main occupations in which “treated” individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Differences between higher- and lower-educated individuals seem to emerge only in later periods and only for the expected tenure indicator. On average, the expected tenure of lower-educated individuals decreases by approximately 168 days after displacement, whereas the decline is less pronounced for higher-educated individuals, at around 97 days. This substantial difference in post-

²⁵ More precisely, in the never-treated specification, females appear to modestly recover from the negative effects of dismissal over time when looking at the expected tenure indicator, but their recovery is slower-paced, discontinuous and less pronounced than that of males.

treatment averages primarily reflects the recovery observed among higher-educated individuals, who seem to keep recovering from the negative effects of displacement over time, unlike lower-educated ones. In contrast, notable differences cannot be spotted between these two subgroups for the other indicator.²⁶ As for the previously analyzed case, confidence intervals are too wide to assert the presence of statistically significant differences between these two categories. Furthermore, Figure B10 in Appendix B, reporting results obtained with never-treated units, does not clarify whether higher-educated individuals recover faster than lower-educated ones from the negative effects of displacement. Weighting all these elements, these findings provide only a tentative indication that high-skilled individuals may be able to reverse the negative effects of displacement over time. Thus, I conclude that no meaningful differences can be established between educational groups.

Lastly, Figure 6 displays the results from separate regressions with respect to the macro-region of work. Once again, it is not possible to detect statistically significant differences among these three subgroups because of wide confidence intervals. Nevertheless, judging solely from an examination of the point estimates, individuals employed in the North of Italy at the time of dismissal seem to suffer harsher and more persistent consequences from job loss, reporting a rise in unemployment risk of about 2.53 pp and a reduction in expected tenure of about 179 days. An inspection of pre-treatment averages (shown in Table A3), however, reveals that individuals working in the northern regions of Italy are generally employed in jobs with a substantially lower risk of dismissal (about three percentage points less than in the Center and almost five percentage points less than in the South) and a considerably higher expected tenure (about 12% higher than in the Center, and 24% higher than in the South). These pre-treatment differences suggest that individuals working in the North have more to lose from dismissal compared to their counterparts in other regions, potentially explaining, at least partially, the patterns observed in Figure 6. As a matter of fact, a simple back-of-the-envelope calculation (obtained by adding—or subtracting, in the case of the expected tenure indicator—the average treatment effect of displacement to the pre-treatment averages) shows that, although workers in the North experience harsher effects from displacement, the average occupation that they find afterward still carry a substantially higher job security than the average occupation found by workers from the Center and the southern regions of Italy. Moreover, given that unemployment rates in the North of Italy are historically lower than in the rest of the country, these results are consistent with the “stigma effects” theorized in the literature.²⁷

Finally, it is important to highlight that the narrative conveyed by these estimates is not consistent with the one depicted by the estimates retrieved using never-treated units as the control group, as shown in Figure B11 in Appendix B. In this latter case, the evidence regarding heterogeneous treatment effects across different macro-areas in Italy appears, in fact, more mixed.²⁸ Since confidence

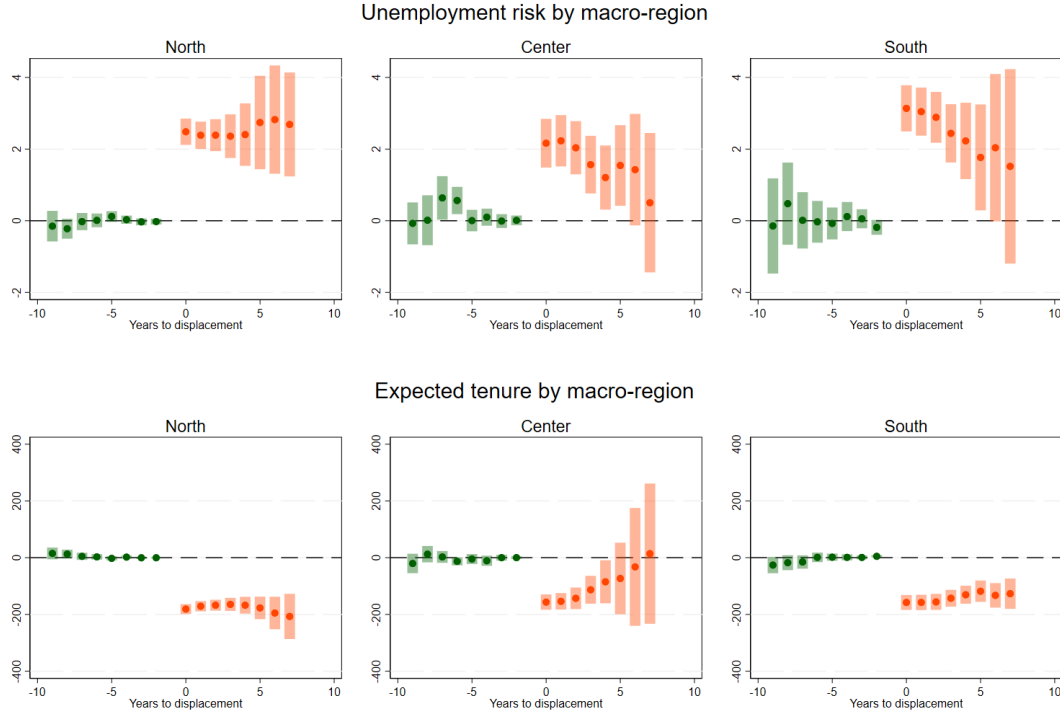
²⁶ Although I observe that the last post-treatment estimate pertaining to highly educated individuals is not significant, this result is likely attributable to limited of statistical power.

²⁷ Omori, 1997, shows that workers who experience “nonemployment” when only few workers are nonemployed are more severely stigmatized.

²⁸ However, notice that, unlike for the previously analyzed cases of education and gender, the region of work might change over time.

intervals do not allow for making definitive statements on the regional heterogeneity of the examined effects, and given the aforementioned inconsistency between the results of the two types of specifications, these findings should be viewed with caution.

Figure 6: Heterogeneous effects of displacement on job security by macro-region



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk associated with the main occupations in which “treated” individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

6 Robustness and placebo

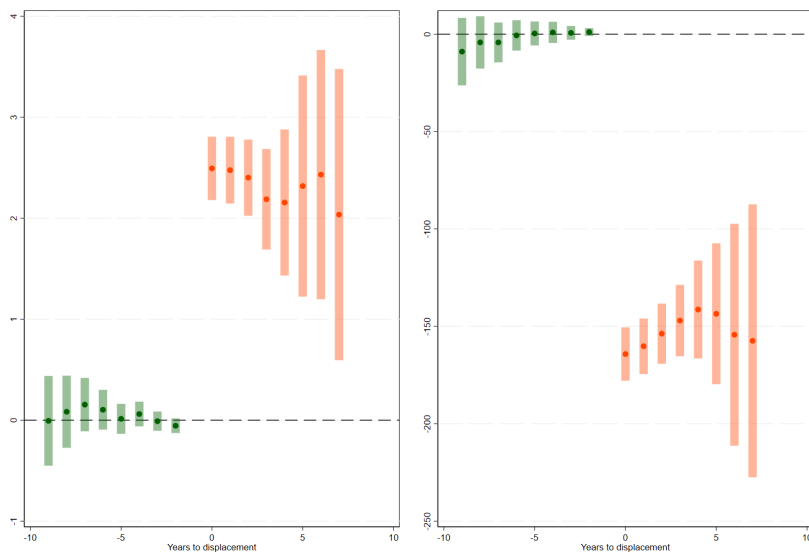
As a final step, I test the soundness of the presented results through a series of robustness checks and sensitivity analyses addressing the following concerns: i) the accuracy of the constructed indicators of occupational job security could be potentially reduced for some sectors and occupations that are scarcely represented in the available sample; ii) the indicators were calculated by including episodes of collective dismissals as events of involuntary job loss, which may raise endogeneity concerns; iii) a small number of outlier observations may be responsible for driving the estimated results; iv) since the main estimates are obtained from a sample of workers aged between 30 and below 54, expanding the sample to include both younger and older worker could yield substantially different results. Finally, as a placebo test, I arbitrarily reassigned the treatment three years prior to its actual occurrence to verify that a significant variation in the outcome variable happened in correspondence

with the timing of the treatment. I will present these exercises in the following subsections.

6.1 Robustness of occupational job security indicators

The first check addresses concerns that may arise from the fact that some sectors and occupations are underrepresented in the sample, potentially making the computation of job security indicators imprecise. To mitigate these concerns, I excluded from the analysis sectors and occupations with fewer than one hundred available observations per year in the original sample. This way, I retained only those occupations that ensured a fairly precise computation of both outcome measures. Consequently, individuals employed in these excluded occupations at any point during the considered period were also discarded. After these operations, the sample size decreased to 94,653 observations. The estimates resulting from this refined sample are plotted in Figure 7, closely resembling those from the main specification and differing only slightly in magnitude.

Figure 7: Robustness: excluding occupations and sectors with less than one hundred observations per year in the original sample

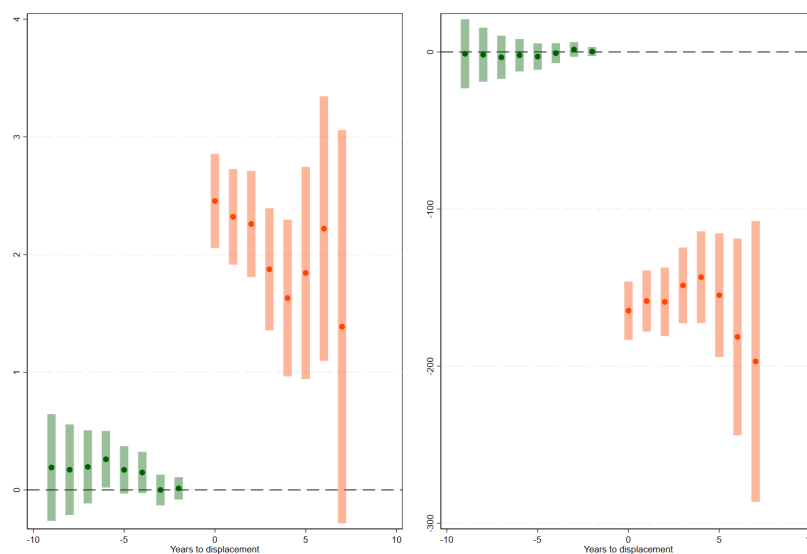


Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (left) and the expected tenure (right) associated with the main occupations in which “treated” individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by [Callaway and Sant’Anna \(2021\)](#).

A second source of concern might stem from the inclusion of collective dismissals in the calculation of the proposed occupational job security indicators, as this may induce endogeneity issues. Two main considerations motivated this choice. First, excluding these events would have misrepresented the inherent security of the affected occupations. Second, these are the occupations workers leave behind when they get dismissed. Since incorporating collective dismissals into the calculation of job security can only make these occupations less secure (regardless of workers’ following occupations),

this approach is only likely to induce a downward bias in our coefficients of interest. Nevertheless, to address this concern, I implemented the following procedure. First, I randomly split the original sample into two groups and re-computed the job security indicators using only one of them. Then, I re-estimated the results using the subset of treated individuals from the group that was not used for the computation of the indicators. This subset consists of 53,885 observations, approximately half the sample used for the main results. The outcome of this exercise is shown in Figure 8. Due to the shrinkage in the sample size, these estimates are less precise than the main results. Still, the magnitudes of the point estimates are comparable: the unemployment risk indicator exhibits a sharper decline over time, whereas the expected tenure indicator appears to move in the opposite direction, suggesting that occupational job security decreases over time. Overall, these findings confirm that the main results remain robust even when the job security indicators and estimates are derived from two disjoint sets of observations.

Figure 8: Robustness: split sample



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (left) and the expected tenure (right) associated with the main treatment in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by [Callaway and Sant'Anna \(2021\)](#).

Lastly, a further source of concern might be related to the use of collective dismissals as a treatment variable. More specifically, it is possible that large-scale layoffs could generate significant spillover effects throughout the local labor market, potentially amplifying the measured effects. For instance, widespread mass layoffs in a particular region might have significant repercussions on regional labor demand, making it more difficult for unemployed workers to find new, stable jobs. Notice that, as already outlined in previous sections, collective dismissals are usually of a smaller scale with respect to the conventional definition of mass layoffs in the literature, which already partially mitigates these

concerns. To further alleviate them, I added regional unemployment rates to the main specification to control for these potential spillovers and re-estimated the model.²⁹ As the CS estimator does not work well with time-varying covariates, I added this control to the regression specification carried out with the methodology introduced by [Cengiz et al. \(2019\)](#). The results of this exercise are remarkably close to those generated by the main specification,³⁰ and are reported in Table A10 in Appendix B.

6.2 Sensitivity of results

One may also worry that outlier observations might be driving the estimates. To confute this concern, I excluded occupations that fall in the top or the bottom 1% of the distributions for either unemployment risk or the expected tenure indicator. Once again, by doing this, I am also excluding all individuals employed in these occupations at any point in time during the sampled period. This operation removes about 3% of the total sample, reducing the sample size to about 104 thousand observations. The estimates, shown in Figure B12 in Appendix B, are similar to those retrieved by the main specification and even slightly higher in magnitude.

For the next sensitivity check, I expanded the sample to include both senior and younger workers, encompassing all individuals aged 20 to 64 years. This operation increased the sample size to approximately 195 thousand observations. The rationale for focusing only on prime-age workers in the main analysis was based on the fact that both young and senior workers might be affected by displacement very differently compared to prime-age workers. Young workers, whose careers are just at the beginning, might suffer more from unemployment because it prevents them from building up the relevant human capital that employers demand. On the other hand, senior workers, being closer to retirement, may face significant challenges in finding new employment, as available late-career opportunities may be limited or require modern competencies that they do not possess. The estimates of this exercise are displayed in Figure B13 in Appendix B. Overall, these estimates corroborate the findings presented throughout this study. However, the unemployment risk indicator behaves quite differently, with treatment effects intensifying in the later periods.

6.3 Placebo test

Finally, as a placebo test, I reallocated the treatment three years prior to its actual occurrence. The specific number of years was arbitrarily chosen and could have been set in any other year preceding the actual treatment.³¹ Figure 9 shows the results originating from this exercise.

The chart on the left depicts a very small (about 0.22 pp on average) yet statistically significant

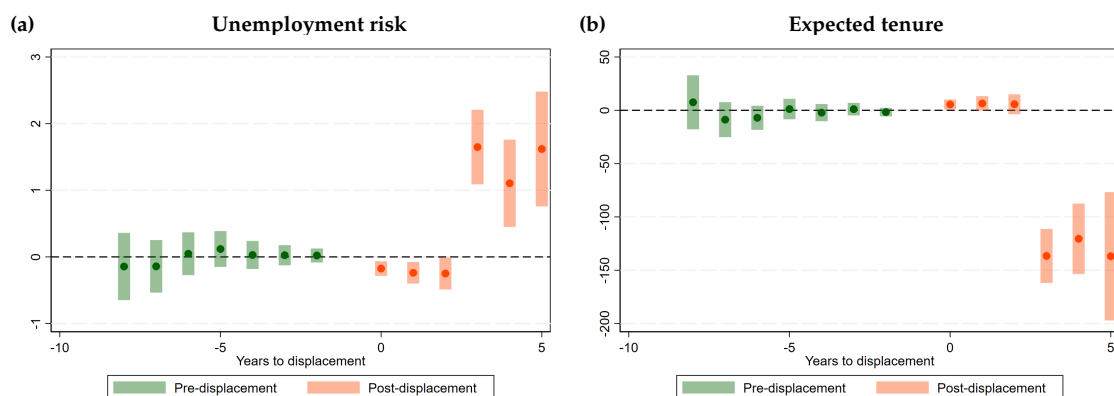
²⁹ This covariate was not included in the main specification because it might represent a "bad control," as it is likely to be endogenous to the treatment variable and potentially bias the estimates. As a matter of fact, local unemployment rates have been frequently used in the literature as instruments for individual layoffs in lieu of collective dismissals (see, for instance, [Gregg, 2001](#)). Nevertheless, for the specific purpose of assessing the incidence of the potential spillovers generated by mass layoffs on the main results, concerns about causality can be set momentarily aside.

³⁰ In this case, the benchmark results are the ones retrieved using the methodology as in [Cengiz et al. \(2019\)](#), reported in Table A7 in Appendix A.

³¹ Indeed, I replicated this placebo test, retiming the treatment to two years before the actual one. The results of this exercise are shown in Figure B14 in Appendix B.

negative effect for the three periods following the placebo treatment. However, while these estimates are all significant, they also are of the opposite sign with respect to the ones pertaining to the true treatment. If anything, this may suggest that individuals, prior to the treatment occurrence, were on a positive trajectory and were gaining over time occupational job security. Moreover, these small negative effects vanish when the placebo treatment is moved, for example, to two years before the real treatment (see Figure B14 in Appendix B). Similar reasoning applies to the expected tenure indicator. In both cases, the real effect only emerges three years after the placebo treatment, aligning with the timing of the actual displacement year. These results reinforce the hypothesis that the findings in the previous section do capture a fundamental change occurring in conjunction with the year of the treatment.

Figure 9: Placebo displacement



Note: The chart displays estimates of the average treatment effect of the placebo displacement on the unemployment risk (left) and the expected tenure (right) associated with the main occupations in which “treated” individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by [Callaway and Sant’Anna \(2021\)](#).

7 Conclusions

This paper aims to provide new insights into the literature on the negative effects of job loss by analyzing how displacement affects the job security of post-displacement occupations. For this purpose, I construct two measures of occupational job security and empirically estimate the causal impact of displacement on these outcomes, exploiting collective dismissals as exogenous variations and the staggered treatment timing at which they occur to define treatment and control groups. While previous research has extensively investigated the effects of displacement on workers’ earnings, wages, and future unemployment, this study narrows the focus to the analysis of the characteristics of the jobs into which displaced workers sort after job loss.

Focusing on variation in occupational job security is particularly relevant for two reasons. First,

it quantifies the extent of the downward fall in the occupational job ladder, isolating, as much as possible, demand-side factors in the post-displacement labor market from individual observable and unobservable characteristics. Second, it suggests that different occupations, by carrying different inherent job security, may eventually lead to higher chances of repeated unemployment spells. Taken together, the presented findings hint to the presence of underlying demand-side deficiencies in the post-displacement labor market that may contribute to the persistent effects of job loss documented in the literature.

This study documents a significant decline in occupational job security. Post-displacement occupations are associated with an increased unemployment risk of approximately 2.36 percentage points and a decreased expected tenure of about 155 days. The dynamics of the estimated treatment effects on the treated indicate a gradual, albeit modest, recovery from these negative effects over time. While point estimates suggest some heterogeneity across different groups of individuals in these effects, no conclusive evidence can be claimed on the existence of differential effects across individuals of different genders, educational levels, and regions of work.

In light of these findings, consequential policy implications can be drawn. Previous studies on the costs of job loss typically recommend vocational training as a means of mitigating human capital losses and facilitating workers' reintegration into the labor market. However, while vocational training may alleviate such losses, it may not be as effective in tackling other kinds of labor market frictions, for example, from asymmetric information in the job market. Indeed, finding a new occupation requires not only acquiring new skills but also exerting considerable effort in signaling unobservable qualities to potential employers. Combined with poor outside options, this may lead displaced workers to accept occupations with lower job security, resulting in persistently weaker employment prospects.

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A Appendix

Table A1: Distribution of individuals in the treatment group across macro-regions of Italy

Macro-region	N	Percent
North	4822	58.41
Center	1681	20.36
South and Islands	1753	21.23

Table A2: Distribution of individuals in the treatment group across educational levels

Level of education	N	Percent
None or elementary school	172	2.09
Middle school	2353	28.50
High school	4309	52.19
University degree or more	1422	17.22

Table A3: Mean value of the outcome variables in the treatment group across macro-regions of Italy in the treatment group in pre-treatment years

Macro-region	Unemployment risk	Expected tenure
North	16.73 (9.11)	1422.99 (588.92)
Center	19.90 (12.54)	1249.76 (603.83)
South and Islands	21.76 (13.82)	1150.13 (575.30)

Notes: Standard errors in parentheses.

Table A4: Mean value of the outcome variables in the treatment group across educational levels in the treatment group in pre-treatment years

Level of education	Unemployment risk	Expected tenure
None or elementary school	24.54 (14.40)	972.61 (520.21)
Middle school	20.90 (11.60)	1179.17 (603.61)
High school	17.58 (10.62)	1381.29 (583.74)
University degree or more	16.34 (10.90)	1459.00 (589.15)

Notes: Standard errors in parentheses.

Table A5: Mean value of the outcome variables in the treatment group by gender in the treatment group in pre-treatment years

Gender	Unemployment risk	Expected tenure
Male	18.55 (11.07)	1331.78 (611.65)
Female	18.19 (11.42)	1327.89 (576.94)

Notes: Standard errors in parentheses.

Table A6: Distribution of observations across cohorts of treated individuals

Year of treatment	Number of treated ids	Percentage
2013	1165	14.11
2014	1459	17.67
2015	1742	21.10
2016	1218	14.75
2017	915	11.08
2018	539	6.53
2019	411	4.98
2020	367	4.45
2021	204	2.47
2022	236	2.86
Total	8256	100.00

Table A7: Results: Unconditional regression, stacking à la [Cengiz et al. \(2019\)](#) - control group: not-yet-treated units

	Unemployment risk	Expected tenure
Tm11	-0.047	8.034
Tm10	-0.210	0.102
Tm9	-0.195	7.948
Tm8	-0.115	12.799*
Tm7	0.237	-1.832
Tm6	0.250**	-5.975
Tm5	0.058	-1.573
Tm4	0.010	0.644
Tm3	0.010	-0.014
Tm2	-0.024	0.223
Tp0	2.545***	-169.938***
Tp1	2.419***	-161.449***
Tp2	2.282***	-155.371***
Tp3	2.145***	-148.856***
Tp4	2.140***	-144.344***
Tp5	2.243***	-143.003***
Tp6	2.304***	-140.290***
Tp7	2.231***	-136.780***
Tp8	2.268***	-132.166***
Observations	279,924	279,924

Notes: The table above displays estimates obtained via a stacking by event specification à la [Cengiz et al. \(2019\)](#). Stars indicate p-values, with: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Results: conditional regression à la [Callaway and Sant'Anna \(2021\)](#) - control group: not-yet-treated units

	Unemployment risk	Expected tenure
Tm9	-0.083	-2.709
Tm8	0.011	2.472
Tm7	0.146	-3.406
Tm6	0.139	-2.039
Tm5	0.071	-2.687
Tm4	0.071	-1.188
Tm3	-0.004	-0.264
Tm2	-0.047	0.713
Tp0	2.559***	-171.188***
Tp1	2.510***	-165.015***
Tp2	2.458***	-160.230***
Tp3	2.254***	-149.741***
Tp4	2.171***	-141.890***
Tp5	2.397***	-144.877***
Tp6	2.431***	-153.535***
Tp7	2.082***	-156.790***
Observations	107,328	107,328

Notes: The table reports the point estimates displayed in Figures 2 and 3 relative to the specifications run with not-yet-treated individuals as control units.

Table A9: Results: conditional regression à la [Callaway and Sant'Anna \(2021\)](#) - control group: never-treated units

	Unemployment risk	Expected tenure
Tm9	0.461*	-23.089**
Tm8	0.056	-8.566
Tm7	0.189	-9.842**
Tm6	0.135*	-7.633**
Tm5	0.051	-5.449**
Tm4	0.067	-3.123*
Tm3	0.036	-2.384**
Tm2	-0.002	-1.354**
Tp0	2.524***	-165.811***
Tp1	2.329***	-156.970***
Tp2	2.184***	-152.638***
Tp3	2.029***	-147.446***
Tp4	1.847***	-136.975***
Tp5	1.787***	-130.334***
Tp6	2.003***	-134.417***
Tp7	2.073***	-138.913***
Observations	3,364,361	3,364,361

Notes: The table reports the point estimates displayed in Figures B4 and B5 relative to the specifications run with never-treated individuals as control units.

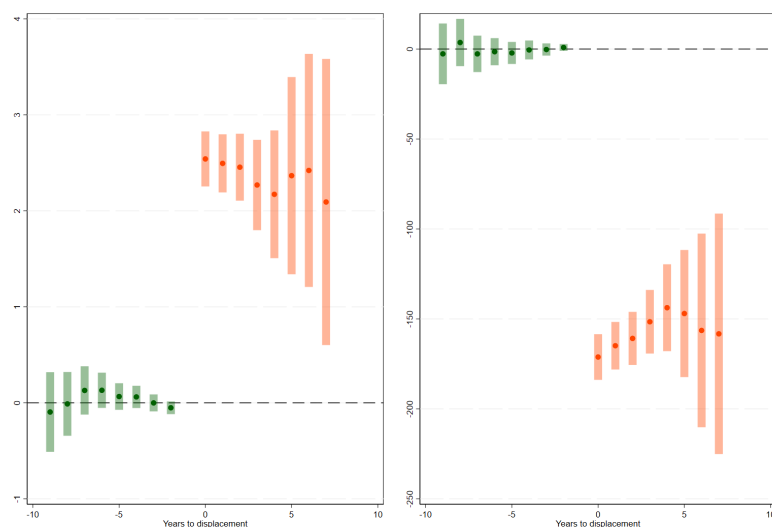
Table A10: Robustness, spillover effects on regional unemployment rates: Unconditional regression, stacking à la [Cengiz et al. \(2019\)](#) - control group: not-yet-treated units

	Unemployment risk	Expected tenure
Tm11	-0.072	8.490
Tm10	-0.221	0.299
Tm9	-0.197	7.983
Tm8	-0.120	12.887*
Tm7	0.238	-1.851
Tm6	0.249**	-5.960
Tm5	0.056	-1.537
Tm4	0.008	0.686
Tm3	0.013	-0.065
Tm2	-0.020	0.153
Tp0	2.547***	-169.982***
Tp1	2.421***	-161.488***
Tp2	2.284***	-155.414***
Tp3	2.144***	-148.830***
Tp4	2.134***	-144.245***
Tp5	2.235***	-142.853***
Tp6	2.297***	-140.155***
Tp7	2.229***	-136.750***
Tp8	2.273***	-132.252***
Observations	279,924	279,924

Notes: The table above displays estimates obtained via a stacking by event specification à la [Cengiz et al. \(2019\)](#). Stars indicate p-values, with: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

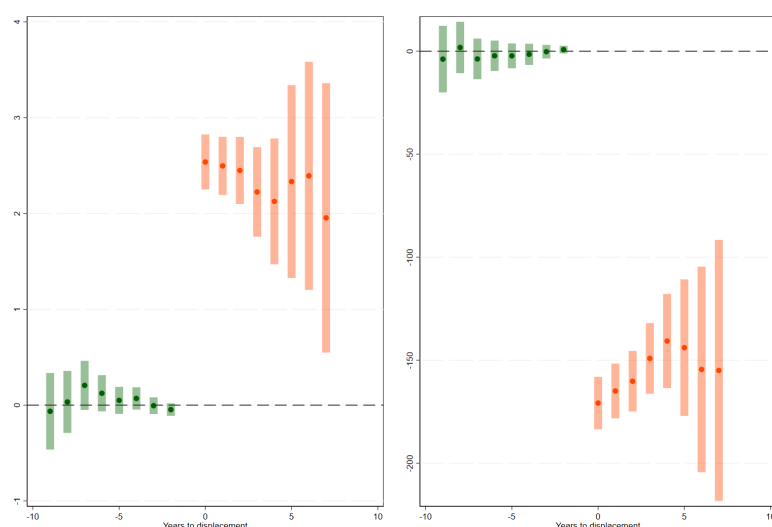
B Appendix

Figure B1: Conditional regression à la Callaway and Sant'Anna (2021) - including workers who lost their jobs more than once due to a collective dismissal procedure



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (left) and the expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by Callaway and Sant'Anna (2021).

Figure B2: Conditional regression à la Callaway and Sant'Anna (2021) - including workers who found their job before the termination date of their previous one



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (left) and the expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by Callaway and Sant'Anna (2021).

Figure B3: Pre vs post-treatment distributions of job security indicators

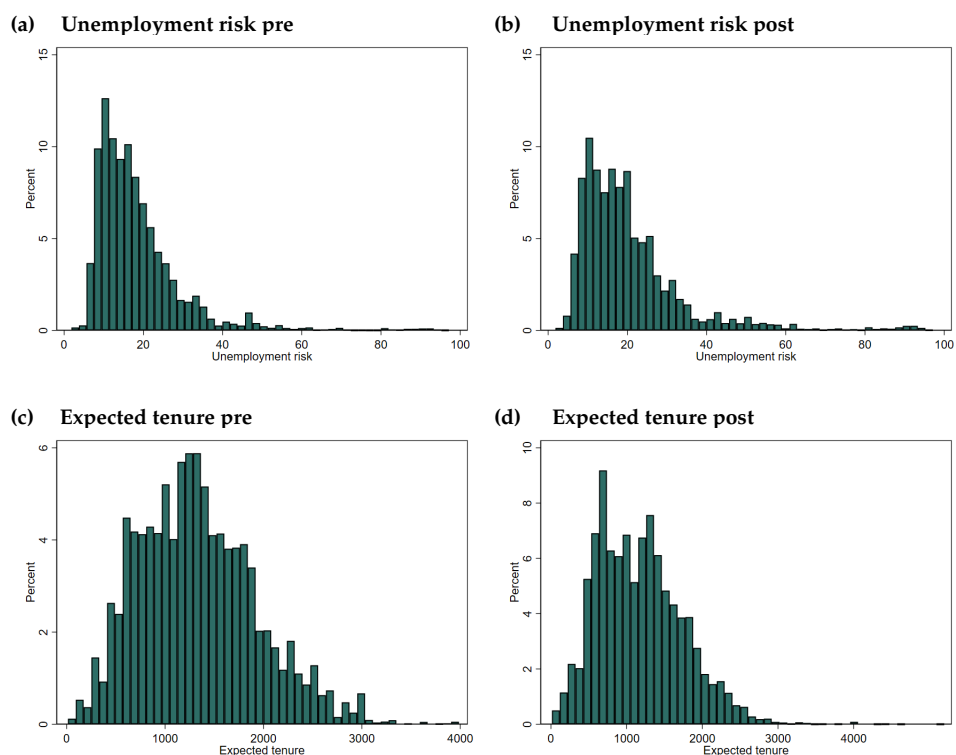
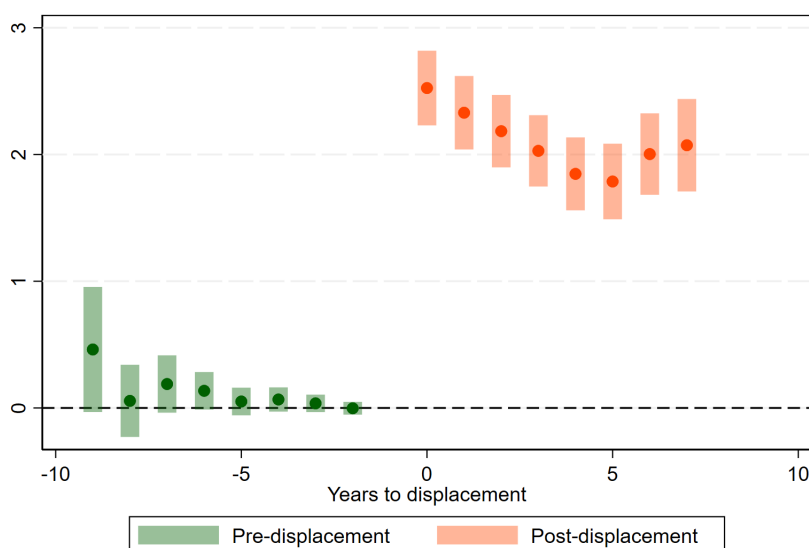
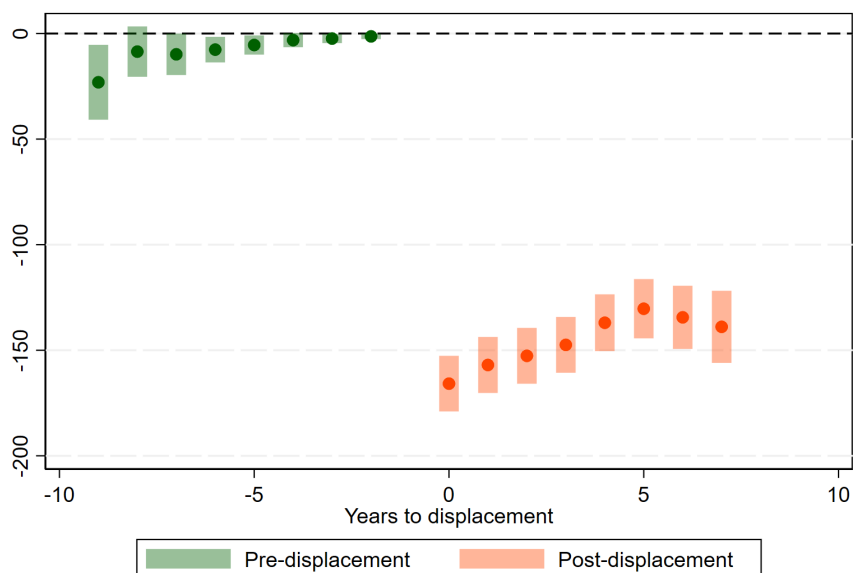


Figure B4: Unemployment risk - control group: never-treated units



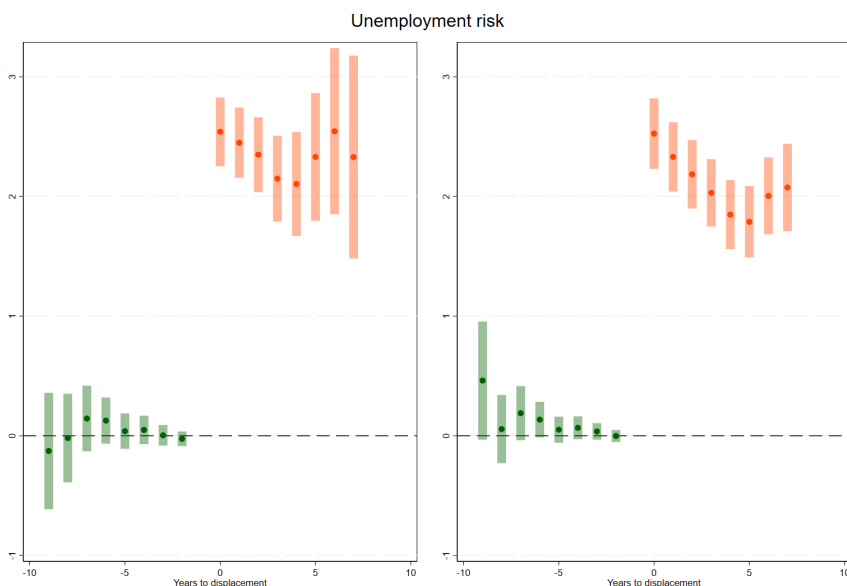
Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by [Callaway and Sant'Anna \(2021\)](#).

Figure B5: Expected tenure - control group: never-treated units



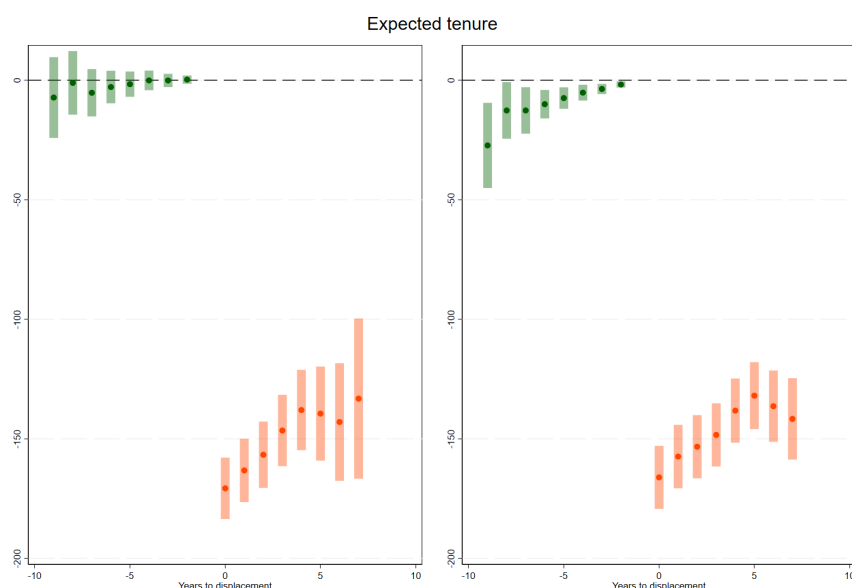
Note: The chart displays estimates of the average treatment effect of displacement on the expected tenure associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by [Callaway and Sant'Anna \(2021\)](#).

Figure B6: Unconditional regression à la [Callaway and Sant'Anna \(2021\)](#) - Unemployment risk



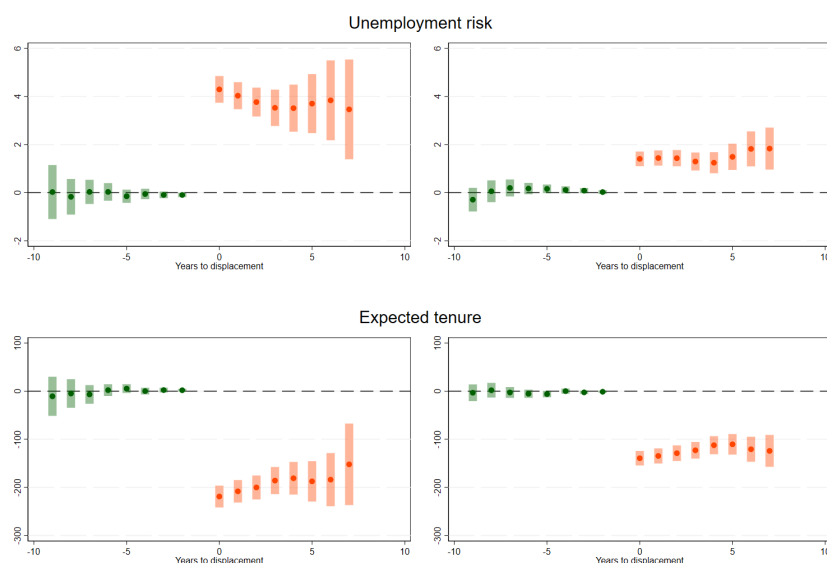
Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk associated with the main occupations in which "treated" individuals are employed each year. Estimates on the left panel are obtained using not-yet-treated units as a control group, whereas estimates on the right are obtained using never-treated units. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B7: Unconditional regression à la Callaway and Sant'Anna (2021) - Expected tenure



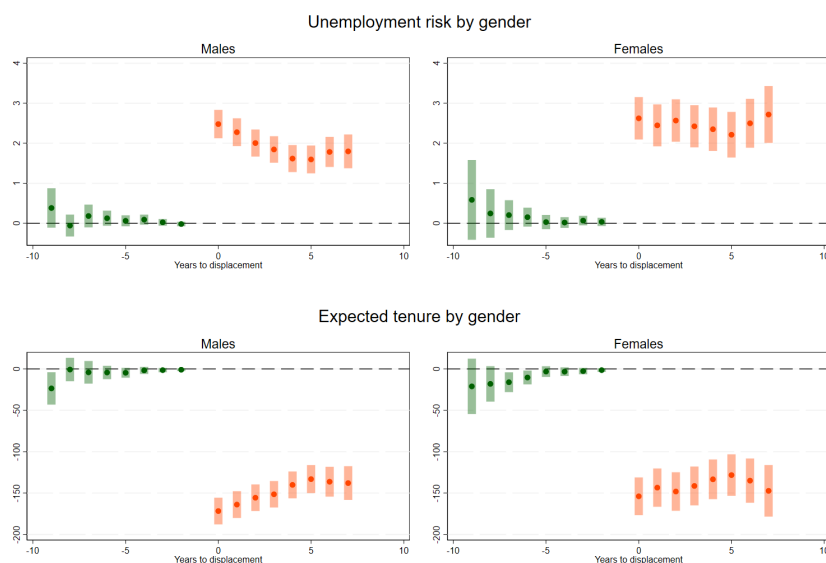
Note: The chart displays estimates of the average treatment effect of displacement on the expected tenure associated with the main occupations in which "treated" individuals are employed each year. Estimates on the left panel are obtained using not-yet-treated units as a control group, whereas estimates on the right are obtained using never-treated units. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B8: Conditional regression à la Callaway and Sant'Anna (2021) by period length between main occupations



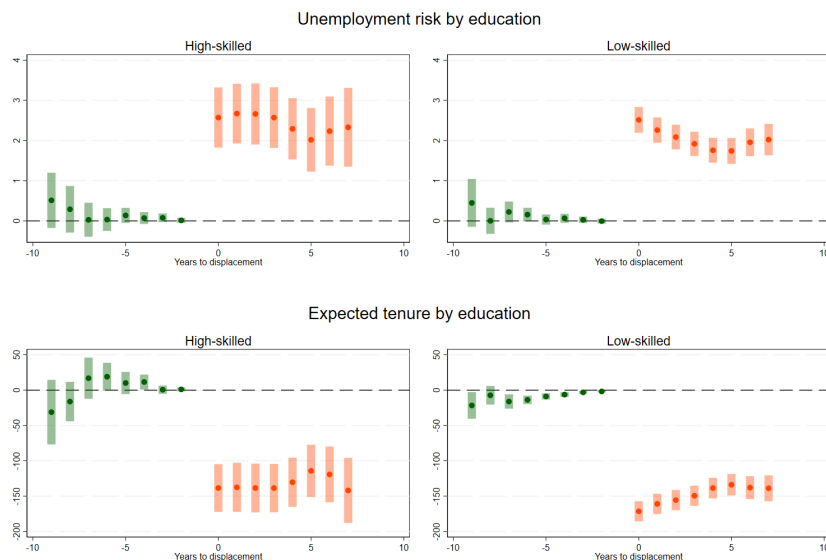
Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (top) and the expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. The charts on the left show estimates obtained for individuals who join their next main occupation following dismissal after a period of six months. The charts' estimates on the right are relative to individuals who joined their next main occupation within a time period of six months. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by Callaway and Sant'Anna (2021).

Figure B9: Conditional regression à la Callaway and Sant'Anna (2021) by gender - control group: never-treated units



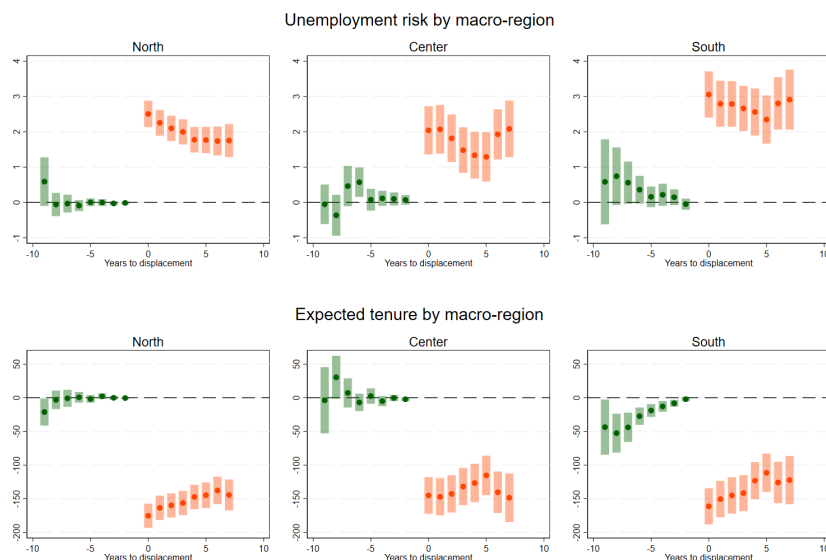
Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (top) and the expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. The charts on the left show estimates obtained for the males in the sample, whereas the charts on the right display estimates for the females. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by Callaway and Sant'Anna (2021).

Figure B10: Conditional regression à la Callaway and Sant'Anna (2021) by educational level - control group: never-treated units



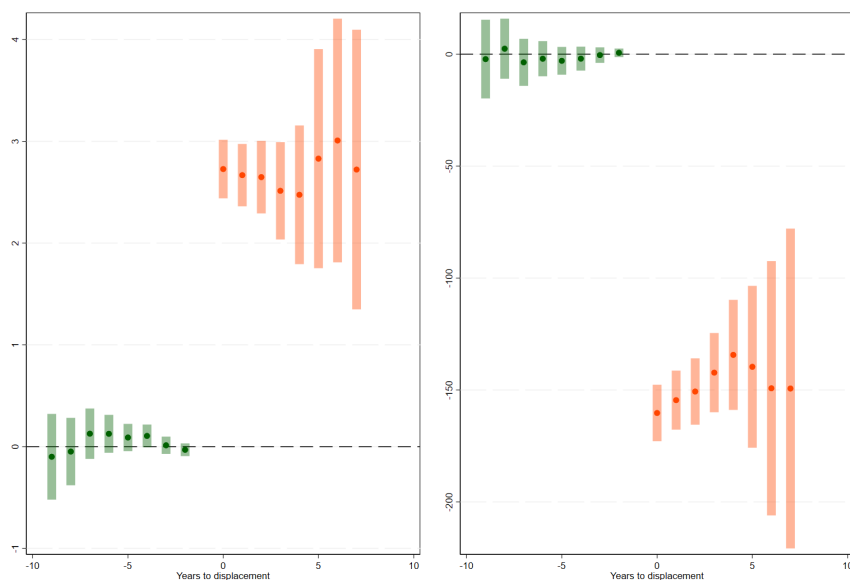
Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (top) and the expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. The charts on the left show estimates obtained for individuals with higher education (having at least a college degree) in the sample, whereas the charts on the right display estimates for the lower educated. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by Callaway and Sant'Anna (2021).

Figure B11: Conditional regression à la [Callaway and Sant'Anna \(2021\)](#) by macro-region of work - control group: never-treated units



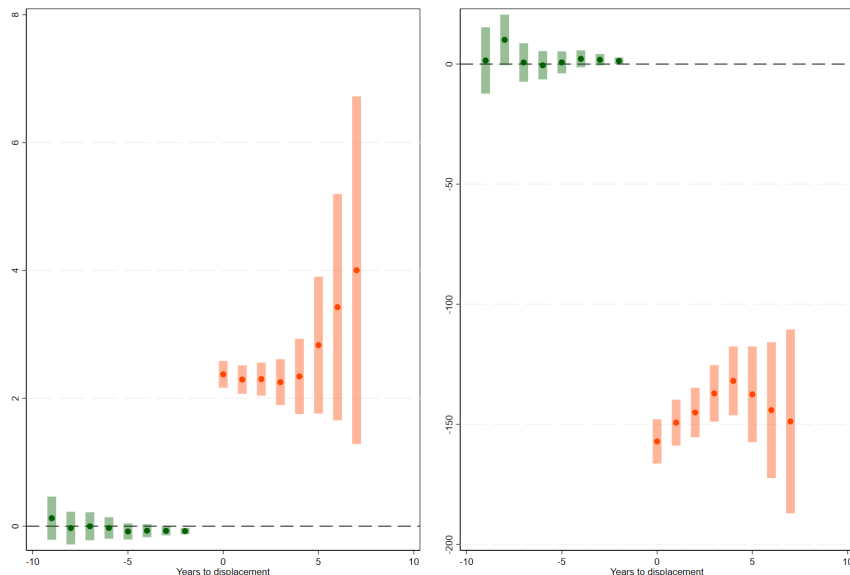
Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (top) and the expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. The charts on the left, the center, and the right show estimates obtained for individuals who worked respectively in the North, Center, and South of Italy at the time of dismissal. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by [Callaway and Sant'Anna \(2021\)](#).

Figure B12: Sensitivity: excluding outlier observations in the bottom and top 1% of the outcome variables' distributions



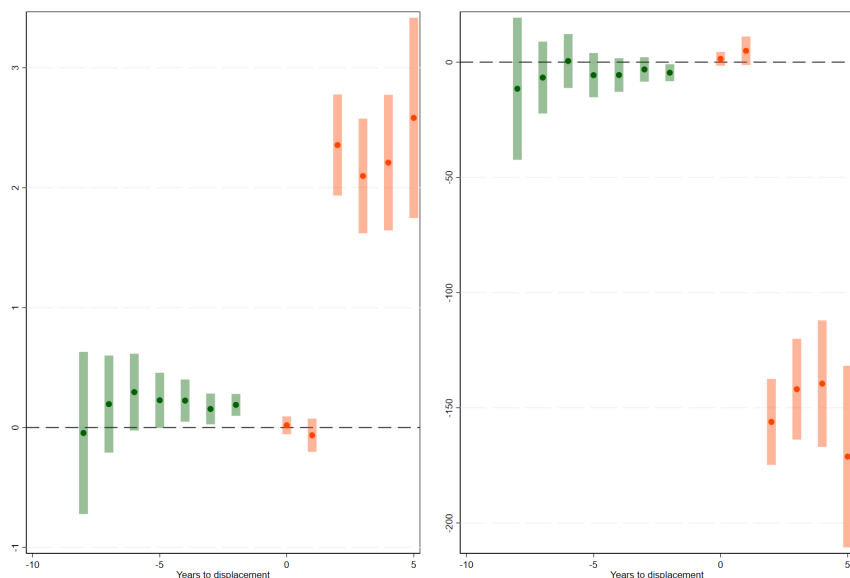
Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (left) and the expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by [Callaway and Sant'Anna \(2021\)](#).

Figure B13: Conditional regression à la Callaway and Sant'Anna (2021) - robustness: including individuals aged 20 to 64



Note: The chart displays estimates of the average treatment effect of displacement on the unemployment risk (left) and the expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by Callaway and Sant'Anna (2021).

Figure B14: Conditional regression à la Callaway and Sant'Anna (2021) - placebo: treatment reallocated two years prior the real one



Note: The chart displays estimates of the average treatment effect of the placebo displacement on the unemployment risk (left) and the expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level. Estimates were retrieved using the methodology suggested by Callaway and Sant'Anna (2021).